CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 10: **The Human Eye and the Colourful World** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. When a second inverted prism is placed in the path of dispersed light, we get 1

(a) white light **(b)** only red **(c)** more colours **(d)** no light

2. The Sun near the horizon appears oval because 1

(a) of scattering **(b)** of diffraction

(c) of atmospheric refraction **(d)** of dispersion

3. In the eye the distance between the lens and retina is nearly constant. So to see objects at different distances the eye varies 1

(a) the size of the retina **(b)** the colour of the iris

(c) the focal length of the lens **(d)** the shape of the cornea

4. A rainbow is seen only when 1

(a) the Sun is behind the observer and water drops are in front

(b) there is a total eclipse

(c) the sky is completely dry

(d) the Sun is in front and drops behind

5. The apparent flickering of stars is more noticeable on 1

(a) a cloudy night

(b) a clear night through a turbulent atmosphere

(c) moon-less night with no air

(d) in space

6. When looking from a close object to a far one, the ciliary muscles 1

(a) contract **(b)** remain tense **(c)** relax **(d)** stop working

7. The colour at the top of a primary rainbow is 1

(a) blue **(b)** violet **(c)** red **(d)** green

8. At high altitude the sky appears 1

(a) white **(b)** red **(c)** darker blue **(d)** lighter blue

9. When one enters a dark theatre from bright sunshine, one cannot see for some time because 1

(a) the pupil takes time to dilate **(b)** the lens fails

(c) the cornea becomes dry **(d)** the retina is damaged

10. The phenomenon responsible for the formation of a rainbow does NOT include 1

(a) internal reflection **(b)** dispersion

(c) refraction **(d)** diffraction by the sun

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do we not see a coloured spectrum with a glass slab? 2

12. Why are danger lights red? 2

13. Why do we get two different focal lengths in a bifocal lens? 2

14. A rainbow is seen when the Sun is behind the observer. Why not when the Sun is in front? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A person is prescribed a lens of power -4 D. Find the far point of the uncorrected eye and the defect. 3

16. Why does the path of a sunbeam entering a dark room through a small hole become visible? 3

17. Explain how raindrops create a rainbow using words: refraction, dispersion, reflection. 3

OR

A person uses spectacles of +2.5 D for reading. Find the focal length and near point of the uncorrected eye (reading at 25 cm). 3

18. Explain why a person cannot see a star in daytime although it is present. 3

OR

A person wearing a -2 D lens has a far point at what distance without glasses? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the structure and working of the human eye as a camera. List three points of comparison with a camera. 5

OR

Why do stars twinkle at night while planets do not? Explain the cause of advance sunrise and delayed sunset, and state the time difference. 5

20. Describe Newton's experiments to show that white light is composed of seven colours. How do you prove that the prism does not add colour? 5

OR

A person needs spectacles of +2 D for reading and -1.5 D for distance. 5

(a) Name the defects

(b) find both focal lengths

(c) state the type of lens to be used.

END OF QUESTION PAPER Total: 40 Marks